```
# importing Necessary Library
```

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
test_data = pd.read_csv('Titanic-Dataset.csv')
```

```
test_data.shape
```

```
(891, 12)
```

```
test_data.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath	female	35.0	1	0	113803	53.1000	C123	S

```
test_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
test_data.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
test_data.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
#droping unnecessary columns which we dont need
```

```
#droping cabin because it is having too much missing values
test_data = test_data.drop(['Cabin'],axis=1)
```

```
test_data = test_data.drop(['PassengerId','Name','Ticket'],axis=1)
```

```
print(test_data.columns)
```

```
Index(['Survived', 'Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare',
       'Embarked'],
      dtype='object')
```

```
# Filling missing 'Age' with median value
test_data['Age'] = test_data['Age'].fillna(test_data['Age'].median())
```

```
# Filling missing 'Embarked' with mode (most common value)
test_data['Embarked'] = test_data['Embarked'].fillna(test_data['Embarked'].mode()[0])
```

```
test_data.isnull().sum()
```

	0
Survived	0
Pclass	0
Sex	0
Age	0
SibSp	0
Parch	0
Fare	0
Embarked	0

dtype: int64

```
test_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 8 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Survived  891 non-null    int64
 1   Pclass    891 non-null    int64
```

```
 2   Sex       891 non-null    object
 3   Age       891 non-null    float64
 4   SibSp     891 non-null    int64
 5   Parch     891 non-null    int64
 6   Fare      891 non-null    float64
 7   Embarked  891 non-null    object
dtypes: float64(2), int64(4), object(2)
memory usage: 55.8+ KB
```

```
#changing Sex column object data type into int so that we can do ml operations  further
mapping = {'male':1,'female':0}
test_data['Sex'] = test_data['Sex'].map(mapping)
```

```
#changing Embarked column object data type into int so that we can do ml operations  further
mapping = {'C': 0, 'Q': 1, 'S': 2}
test_data['Embarked'] = test_data['Embarked'].map(mapping)
```

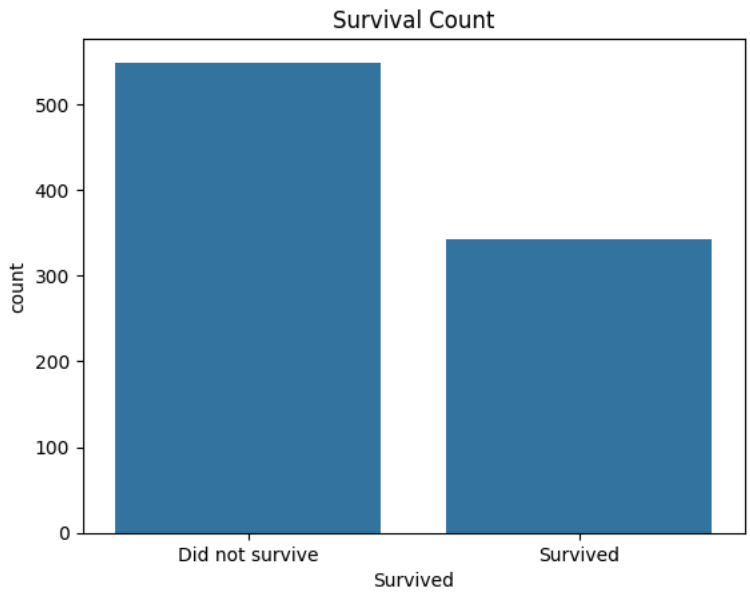
```
test_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 8 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Survived  891 non-null    int64
 1   Pclass    891 non-null    int64
 2   Sex       891 non-null    int64
 3   Age       891 non-null    float64
 4   SibSp     891 non-null    int64
 5   Parch     891 non-null    int64
 6   Fare      891 non-null    float64
 7   Embarked  891 non-null    int64
dtypes: float64(2), int64(6)
memory usage: 55.8 KB
```

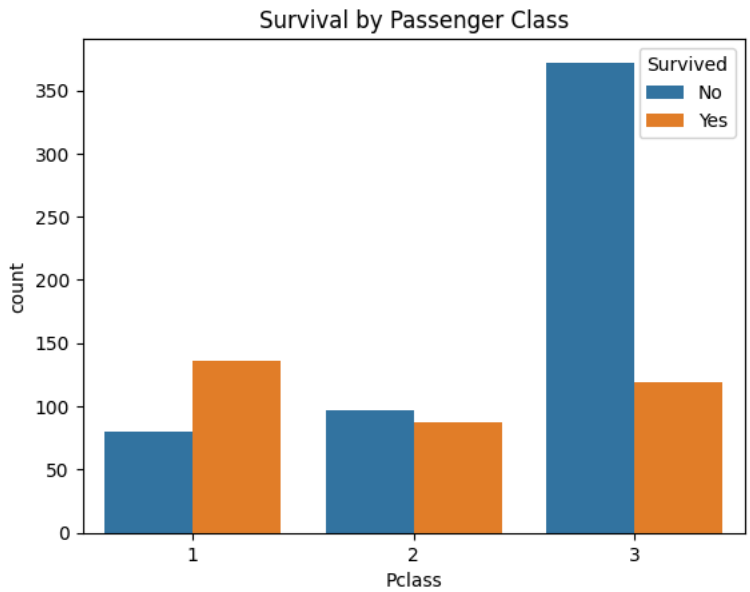
```
test_data.head()
```

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	1	22.0	1	0	7.2500	2
1	1	1	0	38.0	1	0	71.2833	0
2	1	3	0	26.0	0	0	7.9250	2
3	1	1	0	35.0	1	0	53.1000	2
4	0	3	1	35.0	0	0	8.0500	2

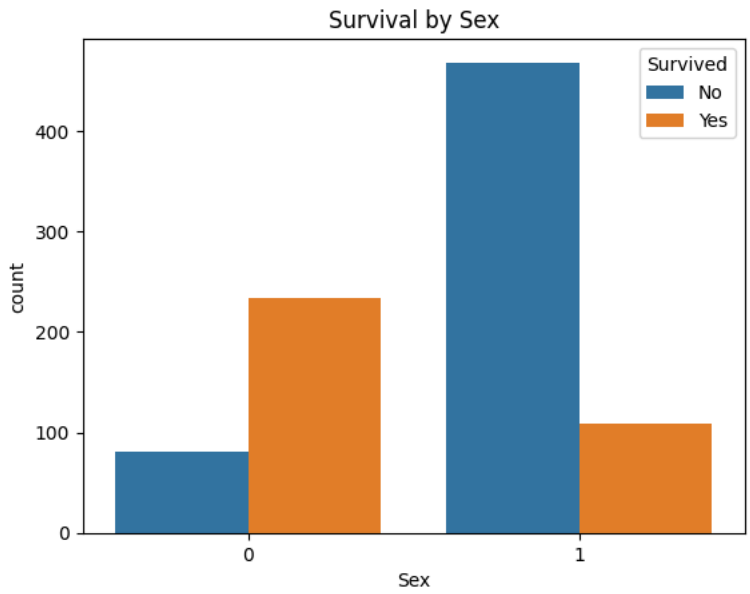
```
sns.countplot(x='Survived', data=test_data)
plt.title('Survival Count')
plt.xticks([0, 1], ['Did not survive', 'Survived'])
plt.show()
```



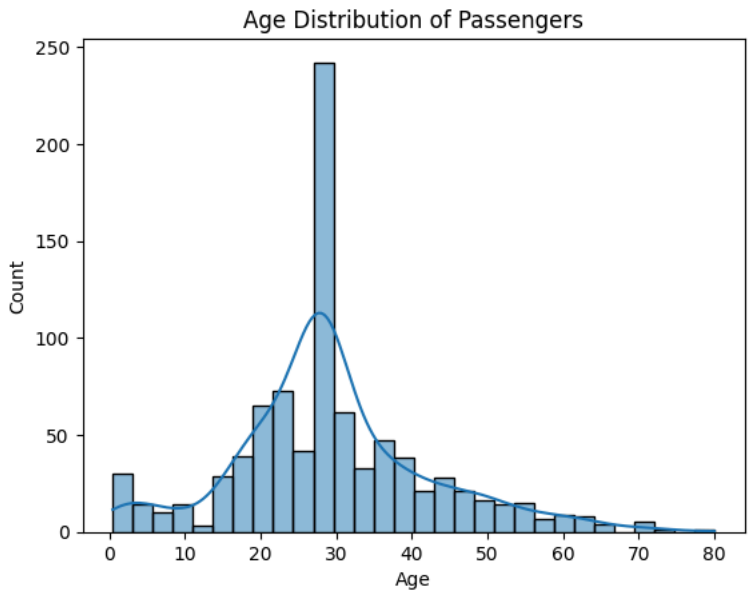

```
sns.countplot(x='Pclass', hue='Survived', data=test_data)
plt.title('Survival by Passenger Class')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()
```



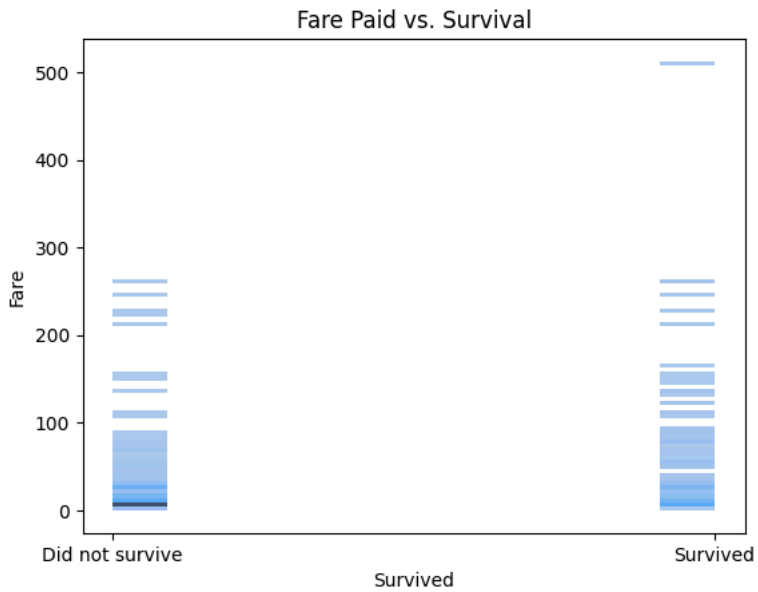

```
sns.countplot(x='Sex', hue='Survived', data=test_data)
plt.title('Survival by Sex')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()
```




```
sns.histplot(test_data['Age'], kde=True, bins=30)
plt.title('Age Distribution of Passengers')
plt.xlabel('Age')
plt.show()
```




```
sns.histplot(x='Survived', y='Fare', data=test_data)
plt.title('Fare Paid vs. Survival')
plt.xticks([0, 1], ['Did not survive', 'Survived'])
plt.show()
```




```
test_data.info()
```

```
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RangeIndex: 891 entries, 0 to 890
Data columns (total 8 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
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 1   Pclass    891 non-null    int64
 2   Sex       891 non-null    int64
 3   Age       891 non-null    float64
 4   SibSp     891 non-null    int64
 5   Parch     891 non-null    int64
 6   Fare      891 non-null    float64
 7   Embarked  891 non-null    int64
dtypes: float64(2), int64(6)
memory usage: 55.8 KB
```

```
#BUILDING A MACHINE LEARNING MODEL
```

```
from sklearn.model_selection import train_test_split
```

```
#Split the Data into Train and Test Sets
```

```
X = test_data.drop('Survived', axis=1)
y = test_data['Survived']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
# IMPORT THE LOGISTIC REGRESSION MODEL FROM THE sklearn.linear_model LIBRARY
```

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
▾ LogisticRegression
LogisticRegression(max_iter=1000)
```

```
y_pred = model.predict(X_test)
```

```
print(y_pred)
```

```
[0 0 0 1 1 1 1 0 1 1 0 0 0 0 0 1 0 1 0 0 0 0 1 0 0 0 0 0 0 1 0 1 1 1 0 0 0
 1 1 0 0 0 0 0 1 0 0 0 0 0 1 1 0 1 0 1 0 1 1 1 0 1 1 0 0 1 0 0 0 1 1 1 1 1
 0 0 1 1 1 0 0 1 1 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 1 0 0 0 1
```

```
 0 1 0 1 0 0 0 1 0 0 1 1 0 0 1 1 1 1 0 1 0 0 1 0 1 1 0 0 1 0 1 0 0 0 1 0 0
 1 0 0 0 0 1 0 0 0 1 1 1 0 0 0 1 0 0 0 1 0 0 1 1 0 1 0 0 0 1 1]
```

```
#Import evaluation metrics from scikit-learn to measure model performance
#accuracy_score: computes the overall accuracy of the model
#confusion_matrix: displays true/false positives and negatives
#classification_report: provides precision, recall, f1-score, and support for each class
```

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.8100558659217877

Classification Report:
               precision    recall  f1-score   support

           0       0.83      0.86      0.84       105
           1       0.79      0.74      0.76        74

    accuracy                           0.81       179
   macro avg       0.81      0.80      0.80       179
weighted avg       0.81      0.81      0.81       179
```

```
#Our model is 81% accurate, which is pretty good.
#It's slightly better at predicting non-survivors (class 0) than survivors (class 1)
```